

Fairness-Aware Bandits for Network Routing in Quantum & Clinical Settings

via Multi-Armed, Contextual, and Informative Contextual Decision Spectra

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Abstract—Sequential decision systems increasingly determine how scarce diagnostic, computational, and communication resources are routed under uncertainty. This study examines fairness-aware bandits across a decision spectrum, from non-contextual multi-armed bandits to contextual and informative contextual bandits, for settings where context is missing, delayed, noisy, or group-dependent. The central claim is that quantum routing and clinical diagnostic routing share a common sequential resource-allocation structure: policies must choose actions from partial feedback, yet groups or flows with the least reliable context can absorb the worst errors.

The work develops a reusable evaluation framework for this bandit spectrum across two executable environment families. The quantum domain builds on a state-preserving quantum-routing evaluation stack, while the clinical domain uses an embedded synthetic clinical environment with resource allocation, clinical models, and EQUITAS-mediated fairness reporting. Preliminary quantum results show that predictive-context policies improve over no-context baselines by 3.74 percentage points internally and 2.83 percentage points externally. In the clinical setting, the embedded EQUITAS-aware allocator improves oracle-normalized efficiency from 75.09% to 91.23%, raises mean success from 50.0% to 72.5%, and reduces the latency parity gap from 18.0 to 10.17 relative to the baseline. In the related bioinformatics proxy, context-aware scoring increases predicted-pair output from 14.96 ± 10.45 to 61.36 ± 44.21 , with optimized output reaching 66.07 ± 46.36 and average disparate-impact improvement of +0.805 after augmentation. Together, these components define a method for measuring and mitigating time-evolving fairness disparities while preserving utility under non-stationarity and limited context.

Index Terms—Fairness-aware bandits, contextual bandits, resource allocation, clinical routing, quantum routing, partial feedback, missing context, algorithmic fairness.

I. INTRODUCTION

Many applied data science systems are sequential decision systems that route limited resources under uncertainty: which diagnostic test to order, which case to escalate, which model or pipeline to trust, or which quantum-network path should receive scarce entanglement-generation attempts. These decisions are difficult because the system observes only partial feedback. The chosen action reveals an outcome, but the unchosen alternatives remain counterfactual. They are also fairness critical because the quality of observed context is often uneven across clinical groups, sites, and cohorts or across quantum flows and service classes. When one population has delayed testing,

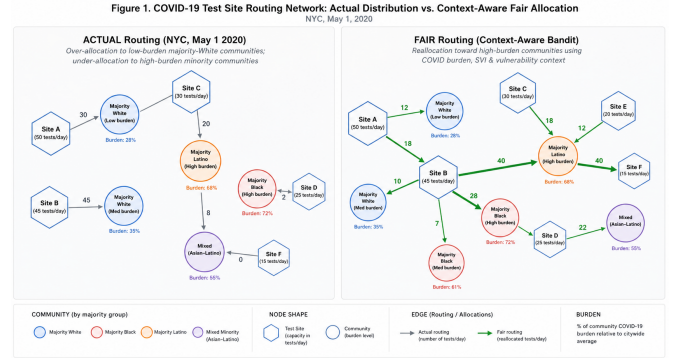


Fig. 1. Real-world clinical routing illustration. The left side shows weak-context allocation, while the right side shows context-aware fairness allocation under the same shared routing problem.

incomplete patient history, lower-quality measurements, or less representative calibration data, a policy optimized only for aggregate reward can hide subgroup errors and compound disparities over time; see Figure 1. The same structure appears in quantum routing when flows depend on uneven link, fidelity, capacity, or route-state evidence; see Figure ??.

This work frames the scarce-resource routing problem as fairness-aware bandit learning across a decision spectrum. At each round, a policy observes available context, chooses an action, receives reward only for the chosen action, and updates its future decisions. The scientific questions focus on whether movement along the bandit spectrum reduces disparities when context is missing, delayed, noisy, or group-dependent. This matters in clinical workflows because false-negative, delayed-escalation, or under-testing gaps can produce direct harm. It also matters in quantum-network routing because future communication systems will allocate scarce probabilistic resources among competing flows; fairness in success probability and latency is a service-quality requirement.

The study develops a shared environment–runner–evaluator architecture for two complementary routing domains. The first is a quantum-routing testbed based on hybrid contextual bandits for entanglement routing and qubit allocation [1]. It exposes noisy links, uncertain entanglement generation,

adversarial or non-stationary conditions, allocator decisions, and saved evaluator state. The second is an embedded clinical simulation motivated by equitable bioinformatics and fairness auditing work [2]. It models test ordering, retesting, escalation, diagnostic attention, model-assisted triage, site-level constraints, and cohort-level context quality. The shared abstraction is that each domain routes scarce resources under partial feedback while fairness risk emerges from unequal context quality.

Four scientific research questions drive this work. RQ1 asks how much quality context improves utility and fairness relative to non-contextual baselines. RQ2 asks how quickly disparities emerge when a group, site, cohort, or flow class receives noisier, delayed, or incomplete context. RQ3 asks whether fairness-mediated policy updates reduce outcome gaps without unacceptable utility loss. RQ4 asks whether the same fairness mediation strategy transfers between clinical diagnostic simulation and quantum routing, or whether mitigation must remain domain-specific. Our preliminary evidence supports the context-gain premise: predictive-context quantum policies improve over no-context baselines by 3.74 percentage points internally and 2.83 percentage points externally. In the embedded clinical setting, the EQUITAS-aware allocator improves oracle-normalized efficiency from 75.09% to 91.23%, raises mean success from 50.0% to 72.5%, and reduces the latency parity gap from 18.0 to 10.17 relative to the baseline.

II. RELATED WORK

A. Bandits and Contextual Sequential Decision-Making

Multi-armed bandits formalize sequential learning under partial feedback: a policy repeatedly chooses among actions, observes reward only for the selected action, and must balance exploration with exploitation. UCB-style algorithms provide finite-time regret guarantees in stochastic settings [3], while Thompson sampling offers a Bayesian alternative with strong empirical performance [4]. The broader bandit literature supplies the regret, uncertainty-estimation, and baseline policy machinery used in this study [5].

Contextual bandits extend this setting by conditioning action selection on observed side information. LinUCB and related linear contextual methods show how context can improve sequential decision quality [6], [7], and reduction-based approaches scale contextual bandits to richer policy classes [8]. However, much of this work optimizes expected reward under the assumption that context is observed accurately, promptly, and representatively. That assumption is fragile in resource-allocation settings where context can be missing, delayed, noisy, or systematically lower quality for particular groups or flows.

Offline policy evaluation is relevant because fairness-aware routing systems often combine logged simulator outputs with fresh online runs. Doubly robust estimators combine reward modeling with inverse-propensity weighting and can reduce variance when evaluating bandit policies from logged data [9]. These methods motivate careful logging of actions, selection rules, context snapshots, outcomes, and oracle comparisons. They also expose a fairness concern: if a subgroup has weaker

support in the logged data, group-specific estimates can remain noisy even when aggregate estimates appear stable.

B. Fairness, Missing Context, and Non-Stationarity

Fairness in supervised learning is often expressed through group-conditioned error criteria such as equality of opportunity and equalized odds [10]. Sequential settings complicate these criteria because fairness constraints affect exploration, data collection, and the future evidence available to the learner. Joseph et al. connect fairness directly to bandit learning [11], while Chen et al. study fair contextual multi-armed bandits with explicit group constraints [12]. Offline contextual-bandit fairness work further shows how importance-weighted estimators can support high-probability constraints [13].

A separate line of work addresses non-stationarity by adapting to changing reward distributions through discounting, change detection, or variation budgets [14], [15]. Restricted- or unobserved-context bandit methods similarly study decision-making when features are unavailable, latent, or costly to observe [16], [17]. These literatures address important parts of the problem, but usually in isolation: standard bandits emphasize regret, fairness work emphasizes disparity constraints, non-stationary bandits emphasize reward drift, and missing-context work emphasizes feature availability. The present study treats missing context, time-varying reward, and group disparity as simultaneous properties of one sequential decision process.

C. Clinical Fairness and Quantum Routing

Clinical algorithmic bias studies show that proxy targets and measurement processes can encode structural inequities, producing under-allocation for some groups even when aggregate performance appears strong [18]. This motivates outcome-based monitoring of false-negative gaps, false-positive gaps, escalation delays, and other group-conditioned harms. In quantum-network routing, decisions are governed by probabilistic entanglement generation, fidelity constraints, storage limits, and scarce network resources [19]–[21]. Existing quantum-routing work primarily emphasizes throughput, fidelity, latency, or robustness. Service equity across flows or user classes remains less developed.

The clinical and quantum settings therefore expose a shared research gap. Both involve sequential allocation under uncertainty, partial feedback, and uneven context quality, but existing methods rarely evaluate outcome-based fairness under missing context and distribution shift across both domains. This gap motivates the shared missing-context routing formulation used in this study.

Table I summarizes how these bodies of work define the technical foundation and remaining gap. Existing methods provide regret guarantees, contextual learning rules, fairness definitions, non-stationary adaptation, and domain-specific routing models, but they rarely connect these pieces into one evaluation framework for fairness-aware routing under missing context. That connection is the role of the proposed clinical–quantum formulation.

TABLE I
LITERATURE AREAS SUPPORTING THE SHARED MISSING-CONTEXT
ROUTING FORMULATION.

Literature area	Role in this study
Bandit foundations	partial feedback, regret, and baseline policies
Contextual bandits	side information and missing-context sensitivity
Fairness definitions	outcome gaps, FNR/FPR, and service equity
Non-stationarity	reward drift and time-varying disparity analysis
Clinical bias studies	subgroup error monitoring and proxy-risk caution
Quantum routing	resource-scarce routing testbed and simulator grounding

III. APPROACH AND METHODOLOGY

A. Shared Missing-Context Routing Model

Both domains are modeled as routing under missing context. The problem formulation identifies the shared harm mechanism: clinical and quantum systems route scarce resources under uncertainty, and unequal context quality can produce repeated allocation harm. Data characterization identifies the measurable patterns that make that harm testable, including group, site, cohort, service line, test distribution, escalation action, latency, reward, success, scenario, allocator, route choice, and evaluator-state fields.

In the clinical domain, the routed resource is diagnostic attention: scarce tests, retests, triage escalation, model-assisted prioritization, or model/pipeline selection. In the quantum domain, the routed resource is entanglement-generation effort, qubit allocation, or path capacity. In both domains, the policy observes partial context, chooses an action, receives partial feedback, and updates. The key experimental variable is the richness and reliability of context available to the policy.

Figure 2 summarizes the shared model. Phase 1 defines the environment and available resources. Phase 2 determines what context is observed and which context is missing or degraded. Phase 3 selects an action using a non-contextual, contextual, or informative contextual bandit policy. Phase 4 evaluates both performance and fairness, then feeds mitigation or updated policy information back into future decisions. This model is useful because it makes the failure point explicit: if context is systematically lower-quality for one group or flow class, the learning process can convert measurement inequity into repeated allocation inequity. It also clarifies where the experiments intervene: missingness is injected before the policy sees the state, while mitigation is evaluated only after utility and disparity have been logged.

At round t , the learner receives an observed context vector $\tilde{\mathbf{x}}_t$, which may differ from the latent complete context $\mathbf{x}_t$ because of missingness, delay, or measurement noise. It chooses an action a_t and observes reward $r_t(a_t)$ only for that action. Utility

is tracked through cumulative regret,

$$R_T = \sum_{t=1}^T (r_t(a_t^*) - r_t(a_t)), \quad (1)$$

where a_t^* is the best available action under the simulator oracle. Fairness is tracked through time-indexed disparity gaps,

$$\Delta_m(t) = |m_t(G=0) - m_t(G=1)|, \quad (2)$$

where m_t is a domain-specific outcome metric such as false-negative rate, success probability, latency, or escalation delay. This notation separates the learning objective from the fairness audit: a policy can have low regret in aggregate while still producing high $\Delta_m(t)$ for specific groups.

B. Bandit Policies and EQUITAS Mediation

The baseline non-contextual policy is an upper-confidence-bound bandit. For action a , let $N_{a,t}$ be the number of times action a has been selected before round t and let $\hat{\mu}_{a,t}$ be its empirical reward mean. The UCB score is

$$U_t^{\text{MAB}}(a) = \hat{\mu}_{a,t-1} + \alpha \sqrt{\frac{2 \log t}{\max(1, N_{a,t-1})}}, \quad (3)$$

and the learner selects $a_t = \arg \max_a U_t^{\text{MAB}}(a)$. This baseline deliberately ignores patient, site, topology, load, or fidelity features, so it measures the cost of acting without context.

The contextual policy replaces scalar action means with action-specific linear models. For each action, let $A_{a,t}$ and $b_{a,t}$ denote the accumulated feature covariance and reward-weighted feature vector, with $\hat{\theta}_{a,t} = A_{a,t}^{-1} b_{a,t}$. Given observed context $\tilde{\mathbf{x}}_t$, the contextual UCB score is

$$U_t^{\text{CMAB}}(a) = \tilde{\mathbf{x}}_t^\top \hat{\theta}_{a,t-1} + \alpha \sqrt{\tilde{\mathbf{x}}_t^\top A_{a,t-1}^{-1} \tilde{\mathbf{x}}_t}. \quad (4)$$

After observing reward $r_t(a_t)$, the selected action updates as $A_{a_t,t} = A_{a_t,t-1} + \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t^\top$ and $b_{a_t,t} = b_{a_t,t-1} + r_t(a_t) \tilde{\mathbf{x}}_t$.

EQUITAS adds fairness mediation as a governed scoring layer around either bandit family. Let $U_t(a)$ be the utility score from Eq. 3 or Eq. 4, $\hat{\Delta}_{m,t}(a)$ be the estimated disparity gap that would result from action a , ϵ be an allowable gap, and $C_t(a)$ be a context-deficit penalty for missing, stale, or low-confidence features. The mediated score is

$$S_t^{\text{EQUITAS}}(a) = U_t(a) - \lambda \max(0, \hat{\Delta}_{m,t}(a) - \epsilon) - \gamma C_t(a), \quad (5)$$

with action $a_t = \arg \max_a S_t^{\text{EQUITAS}}(a)$. The parameters λ and γ control the utility–fairness and utility–context-quality tradeoffs. In clinical routing, $\hat{\Delta}$ tracks error, escalation, or delay gaps across sites and cohorts. In quantum routing, it tracks service gaps such as success probability, latency, or resource access across flows or service classes.

C. Design Decisions

Bandits are used instead of full reinforcement learning because both testbeds provide bandit feedback: only the chosen diagnostic or routing action reveals an outcome. This study compares three policy families. Non-contextual MABs serve

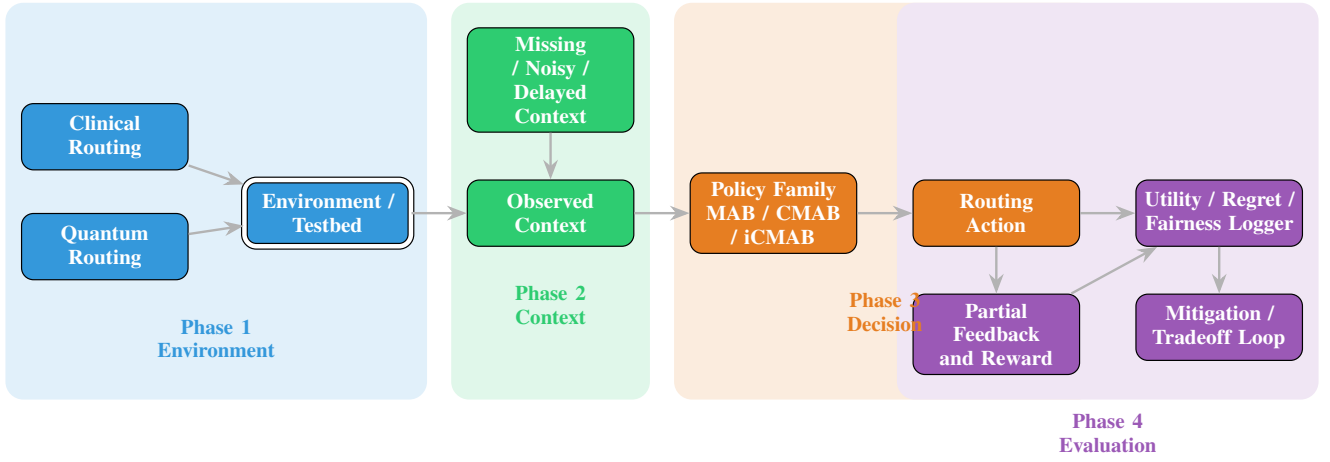


Fig. 2. Shared missing-context routing model. The figure isolates the causal chain that can generate unfair outcomes in both domains: the environment defines the routing problem, missing/noisy/delayed information perturbs only the context stage, the policy family acts on the degraded observation, and utility–fairness logging plus mitigation occur only after partial feedback is received. Clinical actions map to test ordering, retesting, or escalation; quantum actions map to route or allocator decisions.

TABLE II
EQUITAS-MEDIATED FAIRNESS-AWARE ROUTING PROCEDURE.

Input: environment $\mathcal{E}$, action set $\mathcal{A}$, policy family π , fairness metric m , allowable gap ϵ , tradeoff weights λ, γ , horizon T .
1. Initialize per-action bandit statistics, fairness summaries, context-quality summaries, and state provenance.
2. For $t = 1, \dots, T$: observe incomplete context $\tilde{\mathbf{x}}_t$ and context-quality indicators from $\mathcal{E}$.
3. Compute a utility score $U_t(a)$ for each $a \in \mathcal{A}$ using the MAB score in Eq. 3 or the contextual score in Eq. 4.
4. Estimate the group- or flow-conditioned disparity effect $\hat{\Delta}_{m,t}(a)$ and context-deficit penalty $C_t(a)$.
5. Apply EQUITAS mediation with Eq. 5; choose $a_t = \arg \max_a S_t^{\text{EQUITAS}}(a)$.
6. Execute a_t , observe partial reward $r_t(a_t)$ and outcome fields, then update policy statistics only for the selected action.
7. Persist the round state: context, action, reward, outcome, group/flow labels, allocator metadata, EQUITAS score components, and provenance.
8. Return: default utility summaries, fairness summaries, state artifacts, and reporting datasets for clinical or quantum analysis.

TABLE III
APPLICATION MAPPING FOR THE SHARED METHOD.

Element	Clinical setting	Quantum setting
Context	site, cohort, service line, test signals, missingness	topology, load, fidelity, route state, capacity
Action	test, retest, escalation, model-assisted triage	path, route, qubit allocation, retry policy
Reward	diagnostic utility, delay/cost penalty, error outcome	success, latency, fidelity, Oracle-normalized efficiency
Fairness gap	FNR/FPR, escalation delay, access imbalance	success-rate, latency, or service-class imbalance

as a baseline for action-only learning. Contextual bandits test whether observed context improves utility and equity. Informative contextual bandits represent policies that recover, augment, or better weight missing context. The simulation-first clinical design avoids sensitive patient data while allowing controlled missingness and group-specific measurement noise. The quantum testbed reuses an existing, domain-grounded

routing environment so that fairness analysis is built on top of realistic uncertainty rather than a toy network.

Fairness is measured as an online outcome property. In diagnostics, the target metrics are group-wise false-negative and false-positive gaps, along with utility and cost-sensitive reward. In quantum routing, service equity is measured through success-probability and latency gaps across flow groups. Mitigation mechanisms under consideration include fairness-regularized policy updates, group-aware calibration or thresholding, and missingness-aware context augmentation. These mechanisms will be evaluated as tradeoffs rather than as free improvements, because fairness constraints can reduce short-term reward while improving reliability and equity.

D. Planned Mitigation Strategy

The first mitigation family will treat fairness as a penalty or constraint in the policy-update loop. In the simplest form, the learner receives a utility reward and a disparity signal; policy updates are then adjusted when recent group-specific outcomes exceed an allowed gap. This makes fairness operational during learning instead of applying it only after the experiment. A second family will use missingness-aware context augmentation. If a group or flow class systematically receives incomplete

TABLE IV
HIGH-LEVEL IMPLEMENTATION ARCHITECTURE.

Component	Responsibility
Environment abstraction	exposes domain state, actions, rewards, groups, and context quality
Runner	executes model policies over repeated rounds and seeds
Allocator/resource policy	controls scarce test, routing, qubit, or escalation capacity
EQUITAS mediator	applies fairness and context-quality adjustments to policy evidence
Evaluator	aggregates run states into utility, regret, and disparity evidence
State/reporting layer	persists model, runner, evaluator, fairness, and figure artifacts

context, the policy can either acquire additional features, impute missing indicators, or adjust confidence estimates so that uncertainty caused by measurement gaps is not mistaken for low expected value. A third family will test group-aware calibration or thresholding at the decision boundary. This is especially relevant for diagnostic-like escalation, where a small threshold shift can change false-negative rates.

The mitigation comparison will be conservative. The analysis will not claim that lower disparity is automatically better if the policy produces unsafe or unusable utility loss. Instead, each mitigation will be evaluated through utility–fairness tradeoff curves. The preferred policy is expected to be one that preserves most of the utility of contextual learning while reducing time-evolving disparity under missing context. This is why the study compares MAB, CMAB, and iCMAB policies under the same seeds, scenario settings, and logged artifacts: the point is to isolate the effect of context quality and fairness-aware updates rather than confounding algorithm choice with simulator configuration.

IV. IMPLEMENTATION AND REPRODUCIBILITY

A. Architecture

The implementation makes the method in Table II executable without binding it to either domain. As shown in Fig. 3, each experiment is organized around six roles: an allocator, configuration, evaluator, runner, model, and visualizer. The environment supplies domain state, available actions, outcome semantics, and context-quality indicators. The allocator constrains scarce capacity before action selection, the runner executes one or more policy models, and the evaluator aggregates utility, regret, and disparity evidence. EQUITAS sits at the scoring boundary: it can mediate actions during an embedded run or audit saved quantum states without changing the original execution semantics.

Table IV summarizes the component responsibilities. The architecture is domain-neutral but not domain-erasing: quantum experiments instantiate the environment with network topologies, link uncertainty, path decisions, allocator settings, and routing outcomes, while clinical experiments instantiate it with sites, cohorts, service lines, diagnostic actions, resource

constraints, and diagnostic outcomes. Both environments emit state records with comparable context, action, outcome, group, allocator, and EQUITAS fields. That shared state contract supports the context-spectrum quantum results in Fig. 4 and the clinical EQUITAS results in Table XI.

B. State Persistence and Reporting

State is treated as a first-class research artifact rather than a temporary execution byproduct. Model state captures policy-level evidence, runner state captures round-level interactions for a configuration and seed, evaluator state captures cross-run summaries, and reporting state links derived datasets, figures, fairness summaries, and manifests to their source states. This layered record is essential for the preliminary findings: quantum saved states support post-run fairness analysis over existing experiments, while the embedded clinical runner produces state records that can be evaluated with the same fairness contract. As a result, claims about utility, context gain, and disparity can be traced back to the states that generated them.

C. Validation Strategy

Validation is organized around scientific contracts. A valid run must emit non-empty state records with context, action, outcome, group or flow label, allocator, and EQUITAS fields. A valid evaluator summary must preserve utility, oracle gap, success, reward, latency, and disparity metrics. A valid reporting package must link tables, figures, fairness summaries, and manifests to their source states. These checks do not replace scientific evaluation; they protect the evidence chain needed to interpret the preliminary quantum and clinical findings reported below.

V. DATA SETS AND EXPERIMENTAL INPUTS

The current quantum testbed uses simulated quantum-network routing scenarios with probabilistic link success, qubit-capacity constraints, allocator settings, time scales, and stochastic or adversarial conditions. The key outputs are framework-state and model-state artifacts, reward traces, oracle comparisons, winner summaries, efficiency measures, and scenario-level statistics. These artifacts are stored locally or in a shared Drive data lake depending on the execution path.

Table V summarizes the main datasets already available in the quantum framework. The project inherits a validated quantum-routing corpus that already contains scenario-level and experiment-level outputs for four external testbeds and for the internal hybrid framework. These corpora are paired with shared data-lake state artifacts for model and framework persistence. This matters because the fairness-aware analysis can build directly on preserved runner/evaluator artifacts.

A small synthetic validation corpus preserves the shape of the final fairness problem without requiring private data. Each record includes a group label, signal feature, contextual feature, binary outcome, missingness flag, and invalid-record flag. The dataset is deliberately small so controlled validation remains fast, but it exercises the same operations needed by the final project: cleaning missing context, preparing train/test artifacts,

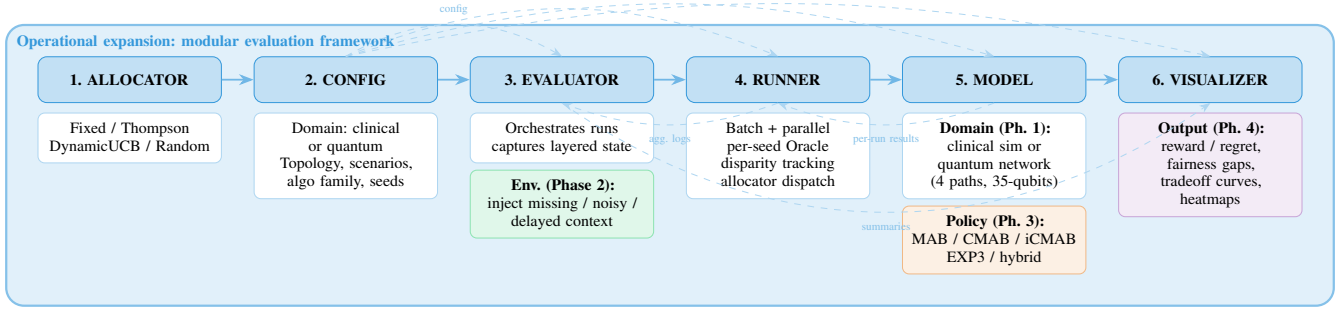


Fig. 3. Operational architecture for the shared method. The stack separates allocator, configuration, evaluator, runner, model, and visualizer responsibilities while preserving the intervention points from Fig. 2: context degradation in the evaluator environment, policy-family choice in the model layer, and utility–fairness tradeoff analysis in the visualizer. This separation is what allows the quantum and clinical findings to be reported through comparable utility and fairness artifacts.

TABLE V
PRIMARY QUANTUM DATASETS AND CORPORA USED BY THE PROJECT.

Corpus	Rows	Cols	Scope	Use in this project
Hybrid master dataset	4,920	30	internal hybrid framework	fairness-layer extension on top of allocator, scenario, reward, regret, and efficiency summaries
Standardized external papers corpus	8,060	32	Papers 2, 7, 8, and 12 at 4K/2K	cross-testbed comparison baseline for adding fairness and transfer analysis
Paper 2 standardized slice	2,450	30	15 nodes, 51 edges, 8 paths	stochastic-noise external benchmark
Paper 7 standardized slice	1,710	30	50 nodes, 141 edges, 15 paths	dense-network external benchmark
Paper 8 standardized slice	1,500	30	20 nodes, 19 edges, 8 paths	compact RL-routing external benchmark
Paper 12 standardized slice	2,400	30	100 nodes, 426 edges, 4 paths	fusion-network external benchmark

TABLE VI
PLANNED DATA FIELDS AND FAIRNESS ROLE.

Field type	Purpose
Group/flow label	subgroup or service class for disparity metrics
Context features	patient/test signals, link/load/fidelity estimates
Missingness flags	degraded, delayed, or unavailable context
Action choice	test, escalation, model, route/allocator decision
Outcome/reward	diagnostic & routing success, latency, or utility
Oracle/baseline	regret, gap & mitigation comparison

producing predictions, computing group metrics, and writing final summaries.

The planned clinical diagnostic testbed will also be simulation-first. Its records will represent patient or case contexts, protected or fairness-relevant groups, diagnostic action choices, observed outcomes, test costs, delay or escalation signals, and injected missingness or measurement noise. This design allows controlled evaluation of false-negative and false-positive gaps before moving to any approved open or sensitive clinical data.

The clinical side also has current preliminary evidence from the EQUITAS bioinformatics project [2], which is used here as a diagnostic-relevant source package. The source package centers on RNA secondary-structure prediction with fairness auditing over sequence type, length, and GC-content groups. Its current corpus contains 20 RNA sequences after fairness-aware augmentation, including tRNA, rRNA, miRNA, synthetic

controls, and 10 targeted Synthetic LowGC sequences added to repair under-representation. These assets provide an existing clinical-context example for method validation while the final sequential diagnostic testbed remains simulation-first.

Table VI summarizes the common schema. The schema is broad enough to support both testbeds but narrow enough to keep analysis comparable: each record has a decision context, a chosen action, an outcome, and a fairness-relevant group or flow label. This design supports cross-domain transfer analysis because the policy-update and mitigation logic can operate over comparable artifact contracts even when the domain-specific feature semantics differ.

The existing quantum corpora already expose many of the fields the fairness layer needs. The master datasets record allocator, scenario, scenario category, run count, frame configuration, capacity semantics, scale, model identity, reward, regret, Oracle-relative efficiency, retries, failures, misses below threshold, and scenario winners. In practical terms, this means the study has access to real experimental data and preliminary results. The main extension still needed is the fairness layer itself: group or flow labels, subgroup disparity summaries, and time-evolving fairness metrics must be added on top of the already preserved performance corpus.

For the clinical side, the project remains simulation-first. The final clinical dataset will mirror the structure of the quantum corpus: context quality, action choice, binary or graded outcome, cost/delay information, and fairness-relevant group annotation. This keeps the cross-domain comparison meaningful. One

TABLE VII
ETHICAL DESIGN REQUIREMENTS AND IMPLEMENTATION RESPONSE.

Requirement	Implementation response
No aggregate-only reporting	utility, subgroup disparity metrics
Expose context quality	log missingness, noise, delay & measurement assumptions
Support accountability	preserve seeds, configs, states, manifests & run provenance
Protect sensitive data	simulation-first data until approved data are available
Run security relevance	quantum routing under unreliable or adversarial conditions

dataset is not meant to imitate the domain semantics of the other; rather, both are designed to satisfy the same artifact contract so that the same evaluation and mitigation logic can be applied to both.

VI. ETHICS, FAIRNESS, PRIVACY, AND SECURITY

Ethics is central to the project because the system is explicitly about resource allocation under uncertainty. A policy that optimizes only aggregate reward can still be ethically unacceptable if it shifts errors onto groups that already have lower-quality context. In the clinical setting, the most direct ethical concern is harm from missed or delayed diagnosis. If context is missing because a group has less reliable testing, delayed follow-up, or lower-quality measurements, then a model can learn that the group is harder to classify and route fewer resources toward it. That is not merely a statistical artifact; it is a mechanism by which historical under-measurement becomes future under-allocation.

The project treats fairness as a design requirement rather than as a post-hoc report. Fairness affects how experiments are built, how artifacts are logged, and how results are interpreted. The evaluation must report subgroup metrics, not only average reward or average efficiency. It must preserve context-quality information, because missingness is part of the causal story of disparity. It must also make assumptions visible: which groups are compared, which outcome gaps matter, which simulator mechanisms generate missingness, and which mitigation thresholds are considered acceptable. Without those details, a reproducible code artifact could still be ethically incomplete because it would hide the value choices that shape the experiment.

Table VII summarizes how ethical concerns translate into design requirements. Privacy and security are secondary analytic goals, but they still shape the project. Clinical data requires privacy-aware handling, which is why the clinical testbed is simulation-first in the current phase. Quantum routing raises a different concern: communication patterns and access to scarce network resources can be sensitive, and adversarial or unreliable conditions can change who receives successful service. The shared framework does not solve every privacy or security problem, but it creates an evaluation structure where

those assumptions can be varied and documented rather than ignored.

The EQUITAS preliminary evidence reinforces the same ethical point: under-represented sequence contexts, such as low-GC examples, required explicit augmentation and re-auditing before fairness claims were defensible. The ethical standard for the final analysis is therefore stronger than “the algorithm works.” The final system should make visible when a policy is accurate but inequitable, robust but unfair to a service class, or fairer only because it sacrifices too much utility to be operationally useful. This tradeoff framing is important for applied data science: fairness is part of the evidence required to justify deployment or to reject a policy as unsafe.

VII. PRELIMINARY RESULTS

A. Quantum Preliminary Results

The preliminary results are organized around the research questions stated in the introduction. The strongest current quantum evidence addresses RQ1: richer context improves routing utility relative to no-context baselines. The quantum side builds on prior threat-aware routing experiments [1], where saved evaluator states, Oracle comparisons, and standardized external corpora already make context-level comparisons possible. These results are not yet full fairness-mitigation results; rather, they identify the utility frontier that EQUITAS must preserve while reducing service disparities.

Figure 4 separates absolute routing quality from the incremental value of added context. The uplift panels show that predictive context is consistently the strongest improvement family, while the size of the gain varies materially by threat regime rather than remaining constant across scenarios.

B. Clinical Preliminary Results

The clinical side contributes preliminary evidence from the EQUITAS bioinformatics source package. This evidence is not presented as final patient-level validation; instead, it supplies a concrete diagnostic-relevant case showing why context availability matters. In this setting, lower context corresponds to structure prediction without thermodynamic/context-aware scoring, while richer context corresponds to Nussinov/MFE feature augmentation, fairness-aware Synthetic LowGC augmentation, and subgroup-aware EQUITAS auditing.

Figure 5 shows that quantum gains are steady and incremental across context levels, whereas the clinical proxy is dominated by a large first-stage recovery when domain-relevant context is restored and then a smaller second-stage gain associated with tuning and fairness-aware augmentation.

Table XIII frames the current evidence conservatively. It does not claim that the fairest policy has already been identified. Instead, it shows that the preliminary results are traceable to saved states, fairness artifacts, and reporting manifests. That traceability is essential for RQ2 and RQ3 because time-evolving disparities and mitigation effects cannot be interpreted from aggregate reward alone.

TABLE VIII
PRELIMINARY QUANTUM RESULTS AVAILABLE BEFORE FAIRNESS-LAYER EXTENSION.

Evidence source	Current quantitative signal	Why it matters	Role in this study
Threat-aware quantum manuscript	pursuit–neural hybrids outperform non-contextual baselines by roughly 18–24 percentage points in scenario-aggregated efficiency	shows that context-rich routing materially improves utility under realistic threats	establishes the performance baseline to which the fairness layer will be added
Threat-aware quantum manuscript	worst-case performance floors remain above 85% under stochastic threats for the strongest hybrid family	suggests the best policies remain stable in benign-but-noisy settings	provides a natural starting point for testing whether fairness can be improved without breaking robust utility
Threat-aware quantum manuscript	replay-capacity increases can trigger 22–30 percentage-point efficiency collapses under adaptive adversaries	shows that resource policy, not only algorithm choice, shapes deployment quality	motivates fairness-aware evaluation of allocator and capacity decisions, not just bandit model choice
Standardized external corpus	8,060 rows across 4 testbeds, 4 allocators, 5 threat scenarios, 3 scales, and 5 routing models including Oracle	demonstrates breadth of already available quantum evidence	supports cross-testbed fairness benchmarking once subgroup/service labels are defined
Internal hybrid corpus	iCPursuitNeuralUCB currently has the highest mean efficiency in the internal hybrid dataset (86.13%)	identifies the strongest current utility baseline in the internal framework	provides the leading candidate for fairness-aware extension and ablation

TABLE IX
CONTEXT-SPECTRUM SUMMARY FROM THE EXISTING GA CORPORA. GAINS ARE MEASURED AGAINST THE NO-CONTEXT FAMILY WITHIN EACH CORPUS.

Context level	Representative model(s)	Internal mean	Δ vs. no ctx.	External mean	Δ vs. no ctx.
No context	GNeuralUCB, EXPNeuralUCB	82.39%	—	62.83%	—
Context	CPursuitNeuralUCB	84.70%	+2.31 pp	64.11%	+1.28 pp
Predictive context	iCPursuitNeuralUCB	86.13%	+3.74 pp	65.66%	+2.83 pp

TABLE X
CROSS-TESTBED PREDICTIVE-CONTEXT UPLIFT IN THE STANDARDIZED EXTERNAL CORPUS.

Testbed	No ctx.	Context	Pred. ctx.	Pred. gain
Paper 2	72.11	73.18	74.34	+2.23
Paper 7	72.87	73.44	78.39	+5.52
Paper 8	70.05	73.69	73.95	+3.90
Paper 12	41.69	42.23	42.54	+0.85

Table VIII makes the quantum evidence concrete. The validated experiment corpus and associated manuscript already provide real model–scenario–allocator–scale evaluations. In the standardized 4K/2K external corpus, the four inherited testbeds contribute 8,060 rows. In the internal hybrid corpus, the strongest current average efficiency belongs to iCPursuitNeuralUCB (86.13%), followed by CPursuitNeuralUCB (84.70%), GNeuralUCB (82.83%), and EXPNeuralUCB (81.95%). These preliminary results identify the current utility frontier and expose which policy families are worth extending with EQUITAS mediation first.

The context-spectrum view in Table IX and Figure 4 directly addresses RQ1. The no-context condition captures reward-only or non-context baselines, the context condition captures methods that use observed contextual structure, and the predictive-context condition captures informative or forecasting-aware extensions. In both internal and external corpora, performance

risers as the method family moves from no context to context to predictive context. The external standardized corpus also shows that gains vary by testbed structure: iCPursuitNeuralUCB reaches approximately 74.34% on Paper 2, 78.39% on Paper 7, 73.95% on Paper 8, and 42.54% on Paper 12. This variation supports the need for cross-testbed fairness analysis rather than a single topology-specific claim.

The clinical evidence addresses the same context hypothesis from a biomedical angle. The EQUITAS package shows that restoring domain-relevant context changes prediction behavior substantially, while under-represented-context augmentation improves parity indicators. These results are preliminary because they come from a clinical proxy rather than the final sequential diagnostic simulator. They are nevertheless methodologically useful: they show that the EQUITAS framing can represent context quality, group representation, utility, and fairness evidence in a form that can later be routed through the same bandit-style evaluator used for quantum experiments.

Together, the preliminary findings support the central novelty of the method. Richer context improves utility, but the magnitude of improvement depends on environment structure; fairness-aware mediation is therefore needed to decide whether context gains are distributed equitably. The next empirical step is to apply Eq. 5 during sequential runs and compare utility, regret, and disparity curves against the no-context and context-only baselines. A mitigation that transfers across both domains would support RQ4; a mitigation that succeeds in one

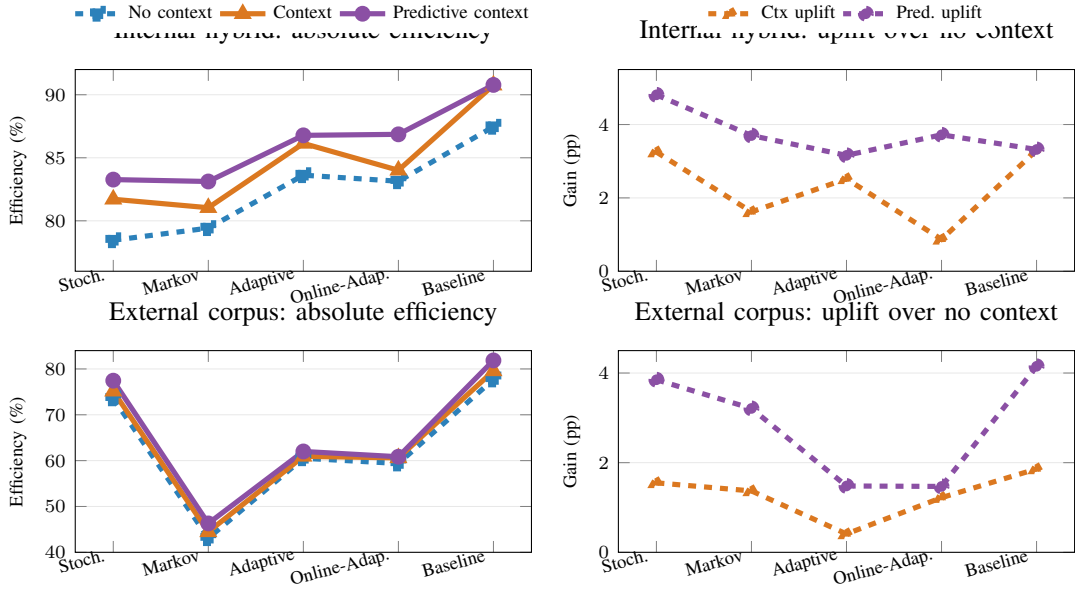


Fig. 4. Quantum context-spectrum results decomposed into absolute performance and marginal uplift. The left column shows the absolute efficiency trajectories for no-context, contextual, and predictive-context families in the internal and external corpora. The right column shows how much contextual and predictive context improve over the corresponding no-context baseline in each scenario. This makes the scientific point explicit: richer context helps everywhere, but the size of the gain is scenario dependent and is usually largest for the predictive-context family.

TABLE XI
PRELIMINARY CLINICAL RESULTS FROM THE EQUITAS BIOINFORMATICS EVIDENCE PACKAGE.

Evidence source	Current quantitative signal	Why it matters	Role in this study
EQUITAS corpus	dataset expanded from 10 to 20 RNA sequences, including 10 Synthetic LowGC sequences	makes clinical-like context gaps measurable in a controlled biomedical proxy task	informs the missingness and group-representation design of the planned simulator
Enhanced Nussinov tuning	Optuna search over 30 trials produced an optimized objective of 56.44 and mean predicted-pair output of 66.07 ± 46.36	shows that richer structural context can materially change diagnostic-relevant prediction behavior	motivates comparing lower-context and richer-context decision policies
Context ablation	thermodynamic/context-aware scoring reported 61.36 ± 44.21 predicted pairs versus 14.96 ± 10.45 without that context	directly supports the context-gain hypothesis in the clinical testbed	anchors the clinical side of the cross-testbed context-gain comparison
Fairness audit	post-augmentation mitigation achieved 6/6 algorithm compliance with the 80% rule, DI ratios of 1.000, and +0.805 average DI improvement	demonstrates that representation/context repair can improve fairness indicators, not only utility	supplies preliminary fairness evidence before final sequential bandit experiments

TABLE XII
AGGREGATE CONTEXT-GAIN COMPARISON ACROSS THE TWO PRELIMINARY EVIDENCE BASES.

Testbed	Lower ctx.	Richer ctx.	Gain
Quantum routing	82.39%	86.13%	+3.74 pp
Clinical bioinformatics	14.96 pairs	66.07 pairs	+51.11 pairs

domain and fails in the other would still be scientifically useful because it would identify which parts of fairness mediation are domain-dependent.

VIII. FUTURE WORK

Next, the focus will be on completing the experimental contribution. First, the quantum fairness adapter will connect evaluator and runner outputs to service-equity metrics such as success-probability and latency gaps. This will turn the existing quantum-routing performance infrastructure into a fairness-aware analysis pipeline. Second, the clinical diagnostic simulator will be implemented with controlled missingness, group-dependent measurement noise, diagnostic action choices, and error outcomes. This simulator will allow the project to test false-negative and false-positive gaps without relying on sensitive patient records during development.

Third, at least one mitigation mechanism will be added and compared against utility-only baselines. The initial priority

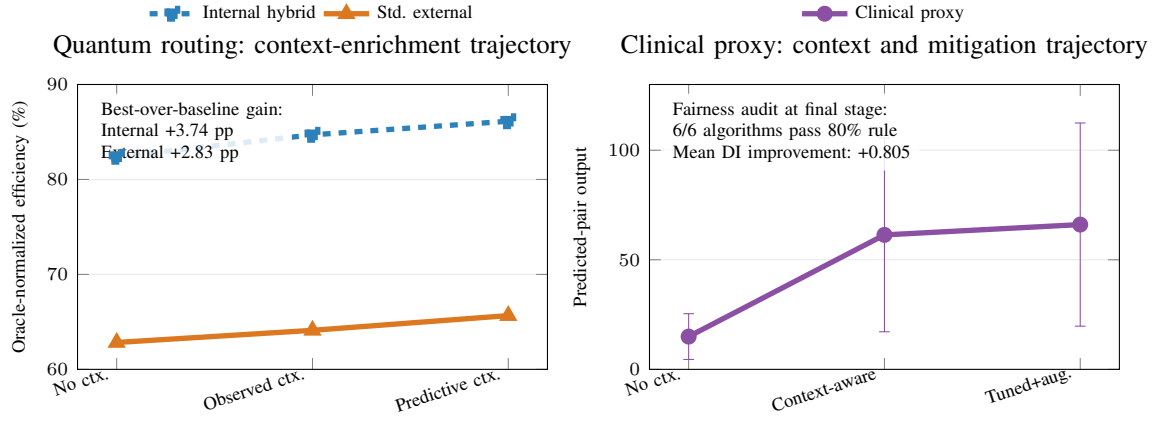


Fig. 5. Cross-testbed context-enrichment trajectories. The left panel shows that quantum-routing performance improves monotonically as the policy moves from no context to observed context to predictive context, with consistent gains in both the internal hybrid corpus and the standardized external corpus. The right panel separates two clinical effects that a single normalized index would obscure: restoring domain-relevant context produces the large primary gain (14.96 to 61.36 predicted pairs), while tuning and fairness-oriented augmentation provide a smaller secondary improvement (66.07 at the final stage) together with improved parity indicators. This decomposition distinguishes pure context recovery from later mitigation or refinement.

TABLE XIII
REPRODUCIBILITY EVIDENCE SUPPORTING PRELIMINARY INTERPRETATION.

Evidence	Interpretation
Quantum state corpora	existing saved runs support context-level utility comparisons
Clinical EQUITAS artifacts	context restoration and augmentation effects are traceable
Fairness sidecar outputs	quantum and clinical states share a governed outcome contract
Reporting manifests	datasets, figures, fairness summaries, and provenance remain linked
Regression tests	state and reporting contracts are protected against implementation drift

is a simple fairness-regularized or missingness-aware update rule that can be implemented in both domains. Fourth, the same policy stack will be evaluated across both domains to test whether fairness mechanisms transfer. Fifth, the final analysis will add full results, plots, statistical discussion, and a stronger conclusion grounded in experimental evidence. The final analysis will also include a limitations section discussing simulation validity, group-definition choices, and the difference between fairness in a controlled testbed and fairness in deployment.

IX. CONCLUSION

This semester established the foundation for a reproducible fairness-aware contextual bandit project. The work completed so far includes a related-work survey, a shared missing-context domain model, an approach and architecture writeup, a cleaned validation package, embedded clinical and quantum execution interfaces, and a governed state-reporting layer. The project is positioned to move from design and infrastructure into final experiments. Its expected impact is a practical framework for measuring and mitigating fairness disparities in sequential

decision systems where context is incomplete, non-stationary, and unequally distributed across groups.

The remaining work is to replace preliminary evidence with complete sequential experiments. The structure needed for those results is now in place: the literature gap is defined, the domain model is explicit, the architecture is reproducible, the data contracts are identified, and the ethical requirements are stated as part of the technical design.

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